

Logging using SLF4J

Exercise 1: Logging Error Messages and Warning Levels

Task: Write a Java application that demonstrates logging error messages and warning levels using SLF4J.

Solution:

```
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-api</artifactId>
  <version>2.0.17</version>
</dependency>
```

PaymentLogger.java:

```
package com.example;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class PaymentLogger {

    private static final Logger logger =
        LoggerFactory.getLogger(PaymentLogger.class);

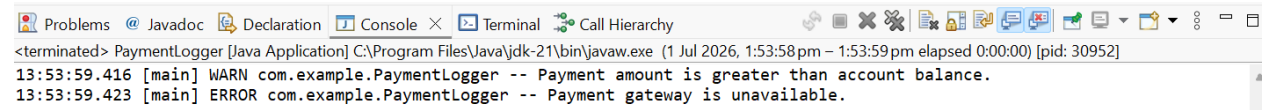
    public static void main(String[] args) {

        double accountBalance = 3000;
        double paymentAmount = 4500;
        boolean paymentGateway = false;

        if (paymentAmount > accountBalance) {
            logger.warn("Payment amount is greater than account balance.");
        }

        if (!paymentGateway) {
            logger.error("Payment gateway is unavailable.");
        }
    }
}
```

Output:



The screenshot shows an IDE interface with a console window. The console title bar includes tabs for Problems, Javadoc, Declaration, Console, Terminal, and Call Hierarchy. The console output shows the termination of a Java application and two log messages: a warning about payment amount and an error about payment gateway availability.

```
<terminated> PaymentLogger [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (1 Jul 2026, 1:53:58 pm - 1:53:59 pm elapsed 0:00:00) [pid: 30952]  
13:53:59.416 [main] WARN com.example.PaymentLogger -- Payment amount is greater than account balance.  
13:53:59.423 [main] ERROR com.example.PaymentLogger -- Payment gateway is unavailable.
```